



AMBER: A Modular Model for Tumor Growth, Vasculature and Radiation Response

Louis V. Kunz¹ · Jesús J. Bosque¹ · Mohammad Nikmaneshi¹ · Ibrahim Chamseddine¹ · Lance L. Munn¹ · Jan Schuemann¹ · Harald Paganetti¹ · Alejandro Bertolet¹ 

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Abstract

Computational models of tumor growth are valuable for simulating the dynamics of cancer progression and treatment responses. In particular, agent-based models (ABMs) tracking individual agents and their interactions are useful for their flexibility and ability to model complex behaviors. However, ABMs have often been confined to small domains or, when scaled up, have neglected crucial aspects like vasculature. Additionally, the integration into tumor ABMs of precise radiation dose calculations using gold-standard Monte Carlo (MC) methods, crucial in contemporary radiotherapy, has been lacking. Here, we introduce AMBER, an Agent-based fraMework for radio-Biological Effects in Radiotherapy that computationally models tumor growth and radiation responses. AMBER is based on a voxelized geometry, enabling realistic simulations at relevant pre-clinical scales by tracking temporally discrete states step-wise. Its hybrid approach, combining traditional ABM techniques with continuous spatiotemporal fields of key microenvironmental factors such as oxygen and vascular endothelial growth factor, facilitates the generation of realistic tortuous vascular trees. Moreover, AMBER is integrated with TOPAS, an MC-based particle transport algorithm that simulates heterogeneous radiation doses. The impact of radiation on tumor dynamics considers the microenvironmental factors that alter radiosensitivity, such as oxygen availability, providing a full coupling between the biological and physical aspects. Our results show that simulations with AMBER yield accurate tumor evolution and radiation treatment outcomes, consistent with established volumetric growth laws and radiobiological understanding. Thus, AMBER emerges as a promising tool for replicating essential features of tumor growth and radiation response, offering a modular design for future expansions to incorporate specific biological traits.

✉ Alejandro Bertolet
abertoletreina@mgh.harvard.edu

¹ Department of Radiation Oncology, Massachusetts General Hospital, Boston, MA, USA

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1 Introduction

Cancer research and medical practice increasingly rely on mathematical and computational models that provide inexpensive quantitative tools for predicting responses and testing hypotheses (McDonald et al. 2023). A paradigmatic example is the calculation of dose in radiation therapy (RT) with the Monte Carlo (MC) method, considered the gold standard (Andreo 2018; Wang and Ma 2013). The idea behind MC simulations is to calculate macroscopic observables by conducting and averaging stochastic simulations of established physical mechanisms at a smaller scale. There is a current trend in using similar methods to implement simple biological interactions and produce simulations of emerging cancer traits (Newton et al. 2013; Reiter et al. 2018; Turajlic et al. 2019; Ortega-Sabater et al. 2022). This kind of approach, incorporating biology into MC and agent-based models (ABMs), provides computational platforms that are used to investigate disease progression (Bosque et al. 2023), explore therapeutic interventions (Zhang et al. 2017; Nikmaneshi et al. 2023; Delobel et al. 2023), and guide pre-clinical and clinical trial design (Grassberger et al. 2019; McClatchy et al. 2020). These methods have permeated the physics realm and impacted our approaches to model biology due to the possibility of calculating the emerging behavior of complex systems from the definition of more elementary mechanistic rules.

ABMs are the best-suited method to model biological systems where multiple factors spatially distributed have interplay effects, such as spatial heterogeneity in tumor growth (Cosgrove et al. 2015; Ghaffarizadeh et al. 2018; Metzcar et al. 2019). ABMs track agents such as cells and blood vessels and impose mechanistic rules that govern how their states change in consecutive discrete time steps according to their current state and those of the surrounding elements. The implemented rules are often stochastic, reflecting the inherently random nature of biological phenomena, such as the specific spatial paths followed by blood vessels (Nikmaneshi et al. 2020; Stepanova et al. 2021) or the occurrence of mutations (Gallaher et al. 2019). Moreover, so-called hybrid ABMs have been developed to integrate continuum-like quantities, such as nutrients or drug concentrations, considered through differential equations (Chamseddine and Rejniak 2020; Nikmaneshi and Firoozabadi 2022). Consequently, given their advantages and flexibility, there is an interest in generating ABMs that can computationally recreate tumor growth and treatment with their spectra of characteristics. Table 1 shows a list of ABMs developed to reproduce tumor growth sorted by type, and marking the presence of three critical elements considered here, namely, tumor growth, vasculature development, and integration of radiation therapy response; in-depth insights on this topic are reviewed elsewhere (Metzcar et al. 2019; Rejniak and Anderson 2010; Roose et al. 2007).

Despite their significant advantages, ABMs for cancer growth and treatment also have limitations. The most significant is their limited scalability regarding tumor volume, as the increasing number of tracked cells leads to increased computational

Table 1 Review chart of published ABMs for tumor growth and angiogenesis

Type of model	Tumor Growth	Vasculature	Radiation	References
2D models of spheroid / small tumors	+	±	–	Drasdo and Höhme (2005), Macklin et al. (2009), Cai et al. (2011), Wu et al. (2013), Kempf et al. (2015)
2D models of spheroid / small tumors with radiation following LQ model	+	–	+	O’Neil (2012), Brüningk et al. (2019)
3D detailed vasculature and tumor growth	+	+	–	Shirinifard et al. (2009), Tang et al. (2014), Cai et al. (2016), Cai et al. (2016), Nikmaneshi et al. (2020), Nikmaneshi et al. (2020), Zangoeei et al. (2021), Nikmaneshi et al. (2020), Nikmaneshi et al. (2023), Duswald et al. (2024)
Vasculature growth algorithm	–	+	–	Norton and Popel (2016), Vavourakis et al. (2017)
3D tumor growth model, based on a mesoscopic ‘large-scale’ method	+	–	–	Jiménez-Sánchez et al. (2021)
Frameworks: Chaste, HAL, PhysiCell, CompuCell, BioDynaMo	+	±	–	Pitt-Francis et al. (2009); Bravo et al. (2020), Ghaffarizadeh et al. (2018), Izaguirre et al. (2004), Breitwieser et al. (2022)
Coupled Frameworks (CompuCell and Geant4) for radiation delivery	+	+	+	Liu et al. (2021)

demands (Randles et al. 2021). This has been partly addressed by mesoscopic models that track cell populations within arbitrary voxels rather than following individual cells (Jiménez-Sánchez et al. 2021; Segura-Collar et al. 2022); however, some of these models lack basic tumor descriptors, such as the presence of vasculature. The presence of blood vessels in the model gains even more relevance when drug and oxygen delivery are considered. Furthermore, the detailed response to treatments, including

their specific sequences and dosages, are often ignored. For the specific case of RT, some models have considered its effect, but usually through simplified considerations such as the linear quadratic (LQ) model, which disregards microscopic stochasticity and biological variability, providing average survival rates to a given dose exposure (O'Neil 2012; Brüningk et al. 2019). The dose and effect of RT can be calculated in detail using measurements and models for radiation-matter interaction. Over the past decade, our group has developed several tools that allow precise calculation of the radiation dose received by tissue in arbitrary scenarios, such as TOPAS and TOPAS-nBio (Perl et al. 2012; Schuemann et al. 2018). The ABM framework allows us to integrate available MC codes of radiation transport and ABMs of tumor growth for a precise description of the effects. To our knowledge, only the work by Liu et al. (2021) has taken this approach, using Geant4 for radiation transport coupled with the ABM CompuCell, but still presenting the critical limitation in the reachable volume previously described.

Here we introduce AMBER (Agent-based fraMework for radioBiological Effects in Radiotherapy). AMBER simulates tumor growth by tracking tumor cells and vascular structure and sets the stage for detailed analyses of responses to RT through its connection with TOPAS and TOPAS-nBio. AMBER is based on an alternative macroscopic approach that discretizes space in voxels instead of tracking individual cells. This allows AMBER to run faster than classical ABMs and reach pre-clinical volumes typical of experimental murine tumors (up to 1000 mm³). Moreover, this structure facilitates comparison with medical imaging such as computerized tomography (CT) or magnetic resonance (MRI), the preferred methods for tumor tracking at macroscopic sizes. AMBER aims to bridge the gap between smaller-scale ABMs and pre-clinical or clinical practice by allowing simulations of relevant sizes, including a representation of the vasculature, and integrating the gold-standard method for radiation calculation. This comprehensive approach can be extended to various treatment modalities, many of which critically depend on spatial effects and vascular status. The current version is a flexible framework that users can extend and adapt to focus on particular problems by incorporating different sets of biological rules relevant to specific settings.

It is expected that this initial development will, in the future, enable the simulation of different radiotherapy modalities and the schedule of their delivery, thus facilitating *in silico* studies tailored to various applications in oncology.

AMBER is designed to contribute to a multi-scale platform to study radiation effects in cancer therapy. This platform spans from the DNA scale (Bertolet et al. 2022) and DNA repair, through levels of nucleus and cell (Schuemann et al. 2018), to the tumor scale (current work), organ level (Huesa-Berral et al. 2023), and whole-patient scale (Shin et al. 2021). This integrated platform will offer a powerful computational system for simulating radiation effects across different scales, enabling further exploration of treatments *in silico* and facilitating patient personalization.

2 Methods: Model Design

2.1 Key Biological Background

Our goal is to model the spatiotemporal growth of vascularized tumors at the macroscopic level. We assume that the tumor is composed of clonogenic cell populations capable of dividing indefinitely, migrating, dying, and interacting with neighboring cells and their environment through a signaling (Hanahan 2022). These cells receive nutrients, e.g., oxygen, from the pre-existing healthy vasculature (Lugano et al. 2020) until nutrient uptake becomes too large or blood vessels are compromised. When that happens, there is a resulting lack of nutrients in the environment, and the cells stimulate angiogenesis to access nutrients from new blood vessels. The uncontrolled proliferation of cells, coupled with the expression or recruitment of extracellular components, causes high intratumoral pressure compared to normal tissues (Nia et al. 2020, 2016). Consequently, elevated intratumoral pressure induces avascular regions as vessels shrink and constrict, leading to areas of hypoxia (Nia et al. 2020; Padera et al. 2004). Hypoxic cells are nutrient-avid and secrete angiogenetic factors, such as vascular endothelial growth factor (VEGF), triggering the formation of new vessels (Shin et al. 2011; Jakobsson et al. 2010). If angiogenesis is insufficient to bring oxygen back to the hypoxic cells, necrotic regions may form in the tumor (Choi et al. 2009; Snyder and Chandel 2009), which often happens at the core of tumors. This necrosis is primarily attributed to the slower clearance of remaining cells by macrophages and other immune cells, disrupting the homeostasis of neighboring cells and exacerbating damage spread. The resulting tumor core becomes particularly resistant to many treatment options due to its inaccessibility and lack of sensitizers such as oxygen (Tang et al. 2014; Forster et al. 2017; Lewin et al. 2018; Suwa et al. 2021).

2.2 Computational Framework Overview

AMBER is a modular framework that can simulate solid tumor growth of pre-clinical sizes, i.e., up to 1000 mm^3 , including hypoxic regions, necrosis, and angiogenesis, as well as the spatiotemporal evolution of the tumor during and after radiotherapy treatment. The AMBER architecture is based on object-oriented programming, where agents and processes are implemented as classes so they can be easily added, modified, or removed. This allows for simple integration of further extensions expanding the biology that is currently implemented. The code is written in Python, a widely known and accessible coding language, and is open source licensed to foster scientific collaboration.¹

The model is designed and implemented with the goal of aligning with the most commonly available data at the relevant scale, which most often comes from CT or MRI scans. To reproduce the format of these data, AMBER's spatial domain is divided into cubic voxels of arbitrary length. This voxel discretization allows us to disregard the individual position of each cell, accelerating computation. For this approximation to be accurate, the voxel has to be small enough to provide a fine discretization of the

¹ <https://github.com/bertoletlab/AMBER>.

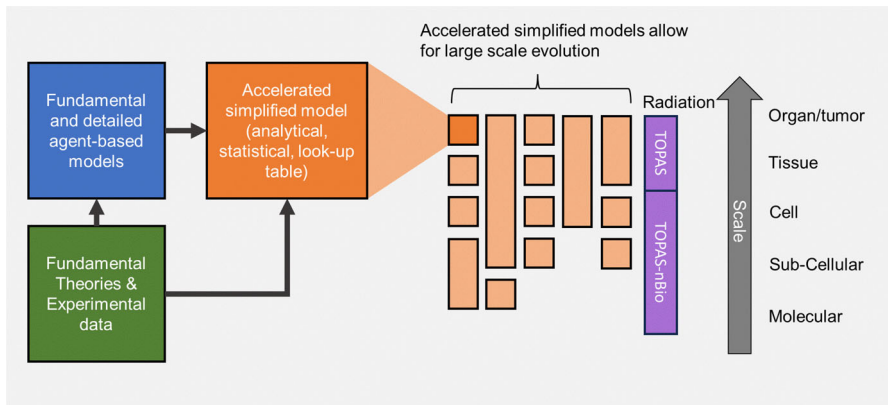


Fig. 1 Schematic of the underlying structure of AMBER. Using fundamental theories and/or experimental data, one can infer accelerated models for a specific process, e.g., oxygen delivery. This inference might need the creation of a custom-made agent-based model, or the remaking/adaptation of a model from literature. The inferred accelerated models can work at one scale or link together different scales and together these models form AMBER. In parallel, radiation can be modelled at all-scales using MC particle transport algorithms like TOPAS and TOPAS-nBio (Color figure online)

tumor and intra-voxel homogeneity of the radiation field yet large enough to ensure that the number of cells within the voxel allows the statistical averaging of the mesoscopic processes. As a consequence, for this work, we set this length to 600 μm .

Certain variables, such as pressure, viscosity, or VEGF concentration, can be adequately represented at the voxel level. Other variables require fine-grained detail of the sub-voxel distributions. This is true for oxygen concentration, which is explained in detail in the following modeling description.

To precisely model these variables, AMBER is structured as a multilayer assembly (Fig. 1) in which custom-made models for specific sub-voxel processes can be integrated to encompass multiple scales. These sub-models can reproduce mathematical expressions, key statistics, or lookup tables that can be applied at a larger scale to replicate their behavior. In parallel, radiation can be modelled at all-scales using MC particle transport algorithms such as TOPAS and TOPAS-nBio (Perl et al. 2012; Schuemann et al. 2018). The current version of AMBER is the proof of concept of a pipeline allowing the simulation of radiation effects from basic principles up to whole tumor scales.

2.3 Implemented Agents and Algorithmic Structure

As shown in Fig. 2, AMBER uses a physical space (“world”) divided into identical cubic **voxels** (600 μm side), each with local properties. Voxels can be filled with **tumor cells**, which can migrate from one voxel to another stochastically. The cells within the voxels are stored as a list, without sub voxel position. Within a voxel, cells can also divide or die. Each voxel is also assigned local, continuous quantities, such as **pressure**, **VEGF** concentration, and **microvessel density**. The pressure is

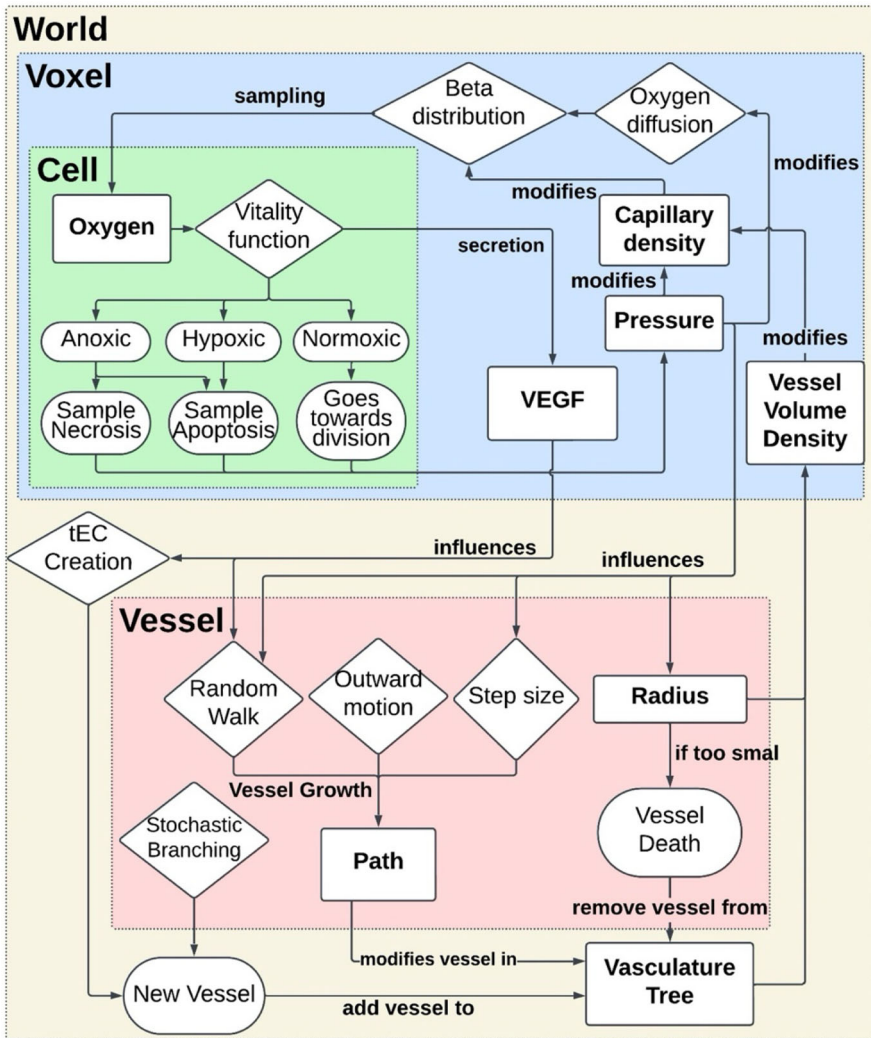


Fig. 2 Structure of AMBER, including the different elements and their interactions. AMBER uses a “world” (beige box on the figure) composed of voxels (blue box) and vessels (red box). Voxels are regular cubic divisions of the space containing cells (green box), and other quantities that can be treated as continuum such as pressure, microvessel density, and molecular factor concentrations (white squares). Inside the green and red boxes, the ‘independent’ processes a cell or a vessel may go through at each time step are illustrated. Characteristics of the cells or vessels may be modified by external elements as symbolized by the arrows entering the green and orange boxes. In turn, cells and vessels can influence other quantities in the voxels such as vascular endothelial growth factor (VEGF) concentration, pressure or vessel volume density as shown by the arrows exiting the green and orange boxes. Steps of the processes or key decisions are shown by rounded squares and diamonds (Color figure online)

the consequence of the space occupied by cells within a voxel, while VEGF can be secreted by the cells depending on the **oxygen level** of the cell. The oxygen level is measured in relative units ranging from 0 to 1, where 0 corresponds to a complete lack of oxygen, and 1 represents the normal physiological oxygen level. This oxygen level in the cell determines the so-called **vitality function** for cells that have not been exposed to radiation. The different levels of vitality define the behavior of the cell and have here been separated into three possible regimes, which are defined by two threshold parameters (see parameter table in SI). These regimes are cycling, which occurs at high levels of the vitality function (corresponding to well-oxygenated cells); quiescent, where cells begin VEGF production to recruit new blood vessels and may also undergo apoptosis stochastically (intermediate levels of vitality); and hypoxic, where cells may die by necrosis due to the extremely low oxygen levels yielding very low vitalities.

The micro vessel density in each voxel is assumed to be a function of the density of the larger **vessels** as micro vessels originate from those. Vessels are discrete structures spanning multiple voxels, defined by their **radius** and a set of points determining their **path**. Each agent is allowed to evolve over time, using discrete steps. The new state of each agent, i.e., a voxel, a cell, or a vessel, is calculated using their states at the previous time step and interactions with other agents, according to a predefined set of rules. Further details about the agents and their actions can be found in Supplementary Materials.

2.4 Vasculature

AMBER simulates the behavior of the vasculature by explicitly tracking vessels with radius larger than 100 μm , while using a microvessel density at each voxel to account for the presence of microvasculature. Each major vessel has a continuous path independent of the voxel discretization used for tracking cells. Each vessel is individually grown and can become occluded or destroyed, influencing the appearance of acute hypoxia within the voxel where it occurs and in all the downstream voxels that the vessel visits. Angiogenesis in the vascular tree occurs by the creation of tip endothelial cells when the VEGF levels are sufficiently high. These tips then move following the combination of the VEGF gradient and a random walk. Vessels can also become occluded and destroyed in response to the high pressure existing in a voxel. See Supplementary Material for further details on the handling of vasculature.

2.5 Oxygen Diffusion Model

As a mean to find an accelerated model as presented in Fig. 1, we look for sub-voxel oxygen distribution. Independently of the macro scale model, we performed repeated voxel simulations at a single voxel scale. We placed n parallel microvessels spreading from one side of the voxel to the other side (Espinoza et al. 2013). Within each voxel we sampled random positions representing the location of cells. For each of these positions we computed the distance to the closest microvessel, and the oxygenation level at that point was calculated according to a logistic function describing the diffusion of oxygen

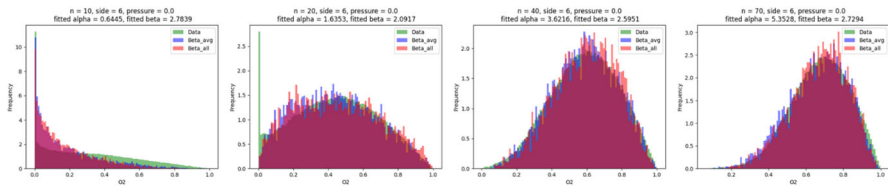


Fig. 3 Beta distribution (blue and red) fitted to the oxygen values (data, in green) generated using the algorithm described in the text for different number of micro vessels (n) and a fixed cell density value (pressure). Here cell density is 0, meaning it is the computation of the oxygen distribution for an isolated cell in the voxel. Resulting parameters α_{dist} and β_{dist} (alpha and beta on the figure) are then stored in a table for the macro scale model (Color figure online)

(see Supplementary Material). We fitted the distribution of oxygen level received by each cancer cell in that voxel using a beta distribution (Fig. 3).

Following this procedure, we simulated a range of voxels changing two key variables: microvascular density n and cell density. With the results of these sub-voxel simulations, i.e., the α_{dist} and β_{dist} parameters resulting from the fit to the beta distribution for each value of the inputs, we built a reference table describing the oxygen distribution of the cells depending on voxel-scale information, i.e., cell density and microvessel density. This reference table is then used by the macro scale model during the tumor growth simulations, speeding up the computation while maintaining the information about heterogeneity at the sub-voxel scale.

2.6 Radiation

We considered the exposure of simulated tumors to ionizing radiation with low linear energy transfer (LET), such as traditional external beam photon radiotherapy. This form of therapy remains the most common in clinical settings, and the low LET allows for dose calculations at the voxel scale, in contrast to higher LET radiation, which would require dose scoring at the cell or even sub-cell scale.

The modeling of the radiation effect was divided into two parts. First, the dose at each voxel for a given arrangement of external radiation was calculated using TOPAS, a specialized Monte Carlo software for radiation transport. Subsequently, the effects on cancer cells and vessels were tracked over time using a damage induction and repair model. In this model, apoptosis and necrosis were triggered with a probability proportional to the existing damage, while damage was gradually reduced at a rate inversely proportional to its current level. In addition to influencing cell behavior, the oxygen level is also considered a modifier of radiation response, as hypoxic cells are more resistant to radiation. Further details can be found in Supplementary Materials.

2.7 Workflow of the Simulation

To initiate a simulation of tumor growth, first, a normal vasculature with a user-definable density is generated using an arbitrary VEGF and pressure map. This

vasculature is designed to replicate the general characteristics of healthy blood vessels, i.e., the vessels run roughly parallel to each other and maintain regular distances between them. Then, a small number of tumor cells (here we used 1000 cells) distributed randomly within a sphere centered at the center of the World and left to evolve freely by performing their defined processes at each time step (divide, migrate, die, etc.). In the current state of the model, healthy pre-existing structures other than the vasculature are disregarded and tumor cells are the only cell population considered. For this work, the voxel size was chosen to be 600 μm , which is in the order of voxel size in a CT scan, and the cell radius was set to 33 μm , representative of a non-small-cell lung cancer cell. The maximum volume ratio of spheres occupying a cube is 0.7, which defines the carrying capacity of a voxel. Therefore, with the chosen voxel and cell sizes, a voxel becomes filled when it reaches a maximum of 1000 cells, which is sufficient for our statistical approach. Above this number, cells cannot enter the voxel through migration, and cells no longer divide.

As cells proliferate and the tumor starts growing, a certain parameter-dependent size is reached where the oxygen supplied by the healthy vasculature is insufficient to maintain the normal functions of the cells. At this point, cells enter a quiescent state and start secreting VEGF, which in turn triggers angiogenesis. The tumor then keeps growing using the newly generated explicit vasculature. Due to the elevated intratumor pressure, especially at the core of the tumor, vessels get compressed or destroyed, leading to the formation of an avascular center. This avascular center creates an anoxic core, which eventually leads to the emergence of a necrotic core. The tumor generated this way could then be irradiated by simulating the radiation fields with TOPAS, causing cell death and tumor shrinkage. Estimated values of the parameters are found in the literature when possible and refined using coarse sensitivity analysis to fit expected large-scale behavior. See Supplementary Table 1 for a detailed list of the parameters used and their values.

2.8 Consistency Testing

To verify that the biological rules defined in our framework result in consistent emergent behavior compatible with the current knowledge, we first checked that a tumor grown with default parameters using the method described above was compatible with well-known tumor growth laws (logistic, Gompertz and Holling), see Table 2 and ref. (Axenie and Kurz 2020) by comparing the trend of these models to AMBER-generated tumor growth rate data. The growth rate equations were then integrated to

Table 2 Equations of the three models considered. N – cell population size, λ – growth rate, θ – death rate, k – carrying capacity (Axenie and Kurz 2020)

Model	Equation
Logistic	$\frac{dN}{dt} = \lambda N - \theta N^2 (1)$
Gompertz	$\frac{dN}{dt} = N(\lambda - \theta \ln N) (2)$
Holling	$\frac{dN}{dt} = \frac{kN}{\lambda + N} - \theta N (3)$

obtain the number of cells as a function of time and compared to the corresponding AMBER-generated data.

Second, we checked that AMBER can reproduce growth rates and volumes observed *in vivo* by modifying the cell cycle duration and the oxygen supply to match the growth to tumor growth murine data (Benzekry et al. 2014; Zou et al. 2020).

Third, we compared the explicit vasculature resulting from tumor growth with default parameters to *in vivo* images of tumors and healthy vasculature (Forster et al. 2017; Seynhaeve and Hagen 2016). Finally, we compared metrics describing the generated vasculature—namely, diameter, vessel length density and vessel segment length—to the values reported in the literature (Nikmaneshi et al. 2020; Norton and Popel 2016; Stamatielos et al. 2014; Gaustad et al. 2017; Kim et al. 2012).

Fourth, our formalism to handle cell radiation response was tested by conducting *in silico* clonogenic survival experiments separately from the AMBER engine. A uniform dose was applied to cells in a single voxel with very high microvessel density to simulate a ‘nutrient-saturated’ environment, matching usual *in vitro* experiments (Franken et al. 2006). Cells were then given sufficient time to fully repair DNA damage with their proliferation switched off. After full repair, the surviving clonogenic cells were counted as alive, excluding those undergoing apoptosis or necrosis due to a poor vitality function. We compared the results for different cell radiosensitivity and repair rates to real experimental data (Jiang et al. 2013) to check whether the macroscopic behavior could be retrieved from the simplified damage induction and repair rules.

Next, a tumor generated using the default parameters and the method described earlier was modeled to be irradiated with 25 fractions, using five sets of five fractions separated by 24 h, and a weekend of pause was left between each set following clinical practices. The irradiations were simulated using the specialized software of radiation transport TOPAS and two opposite 6 MV photon beams were used for this case, even though the method is general and can capture arbitrary configurations. The use of a specialized and flexible code for radiation transport enables the study of arbitrary beam configurations and radiation modalities. Finally, we investigated the effect of cell cycle duration.

3 Results

3.1 AMBER Accurately Reproduces Tumor Evolution and the Emergence of a Necrotic Core

Figure 4 displays a single realization of tumor growth generated by AMBER. The figure shows the emergence of a necrotic core, the modification of the microvessel density map, and the growth of the vasculature through angiogenesis. Initially, the tumor grows along with the pre-existing vasculature (vessel co-option). After that, pressure in the center leads to vascular constriction and hypoxia, resulting in the emergence of necrosis in the tumor’s interior. The microvessel density initially increases when angiogenesis is triggered, and then decreases in the center of the tumor due to pressure. This pattern is also observable in the explicit vasculature depicted in the last row.

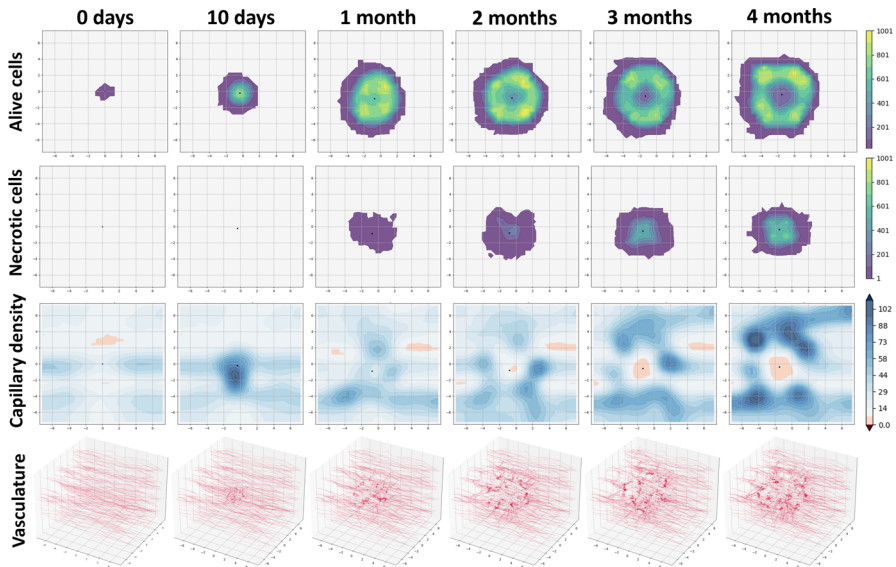


Fig. 4 Evolution of the distribution of alive and necrotic cells in a central slice of a tumor grown with AMBER, along with the evolution of the vasculature, from low density in purple to high density in yellow. Third row shows the micro vessel density, with well oxygenated voxels in blue and hypoxic/anoxic voxels in orange/red. Last row shows the explicit vasculature, with only healthy vasculature in the first column and new vessels growing due to angiogenesis in the next columns (Color figure online)

The left panel of Fig. 5 illustrates the evolution over time of each cell type. The plot differentiates between cycling cells, which have enough oxygen to perform basic functions and are located in voxels that are not full, and quiescent cells, which have stopped proliferation due to lack of nutrients or lack of space within their voxel. It highlights exponential growth until the curves for cycling and quiescent cells intersect. After that point, the forming of new vasculature becomes necessary for the tumor to continue growing, leading to linear growth. Once the vasculature reaches a steady-state, i.e. when the rate of new vessel creation resulting from angiogenesis equals the rate of vessel destruction by pressure, tumor growth slows down until reaching a plateau.

3.2 AMBER Growth Trend Matches Known Growth Laws

The growth of volume over time resulting from AMBER simulations was compared to the logistic, Gompertz, and Holling models. The top right panel of Fig. 5 shows the analytical models and the volumetric data generated by AMBER. The bottom right panel shows the integral of the three fitted models over time, overlaid on the data generated by AMBER. We found an excellent match between the AMBER growth trend and the evolution predicted by the Holling model, and a qualitatively good match with the Gompertz model. However, the logistic model showed significant dissimilarity with our results. The agreement with the first two models indicates that

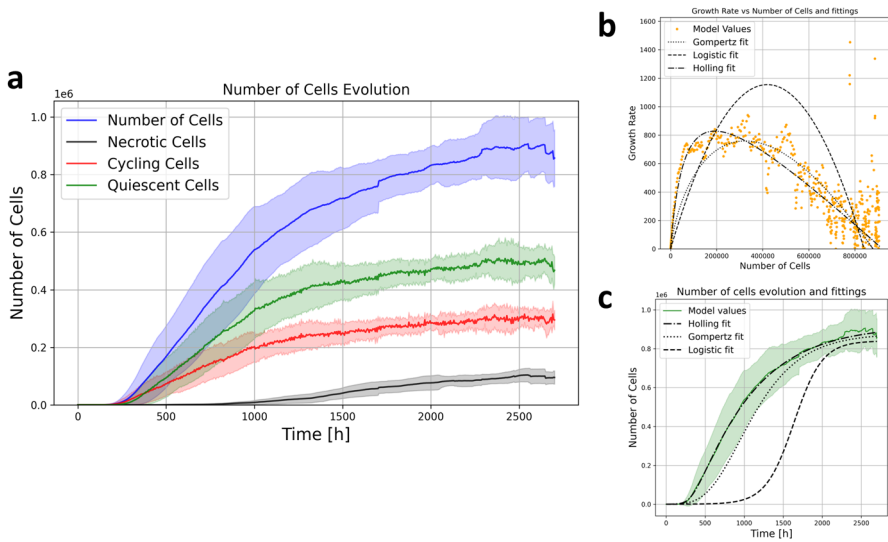


Fig. 5 **a** Time evolution of the total number of cells (blue) and cell groups. The solid line is the average of 15 simulations while the colored area shows plus/minus one standard deviation. **b** Fits of growth rates against number of cells in AMBER simulations to logistic, Gompertz and Holling models. **c** Integral of the fitted equations showing number of cell evolution against time and comparison with the values from the stochastic model. The solid line is the average of 15 simulations while the colored area shows plus/minus one standard deviation (Color figure online)

the time evolution of cell number predicted by AMBER is compatible with well-known laws of tumor growth, while the dissimilarity with the logistic growth indicates that the symmetry in acceleration and deceleration assumed by the latter does not correspond to the conditions implemented here.

3.3 AMBER is able to Replicate Experimental Data Volumetric Growth

By tuning the cell cycle duration and increasing the oxygen supply from the default parameters (see Supplementary Material), the model matches the shape of growth rate and tumor volume data available in the literature of two lung cancer cell lines subcutaneously injected in mice (Fig. 6). Data sets are shifted, so the first time point matches a time point of AMBER.

3.4 The Model Reproduces Vascular Traits of Tumor Angiogenesis

After reaching a critical size angiogenesis is required for the tumor to keep growing. This limiting size depends on the oxygen supply, cell growth rate, and external pressure, all of which are influenced by model parameters, and is around 50 mm^3 with the current parametrization. Figure 7 illustrates the vasculature generated in AMBER during a simulation of tumor growth. On the periphery of the figure, the healthy vasculature generated prior to the tumor initiation remains in its original state in locations far from

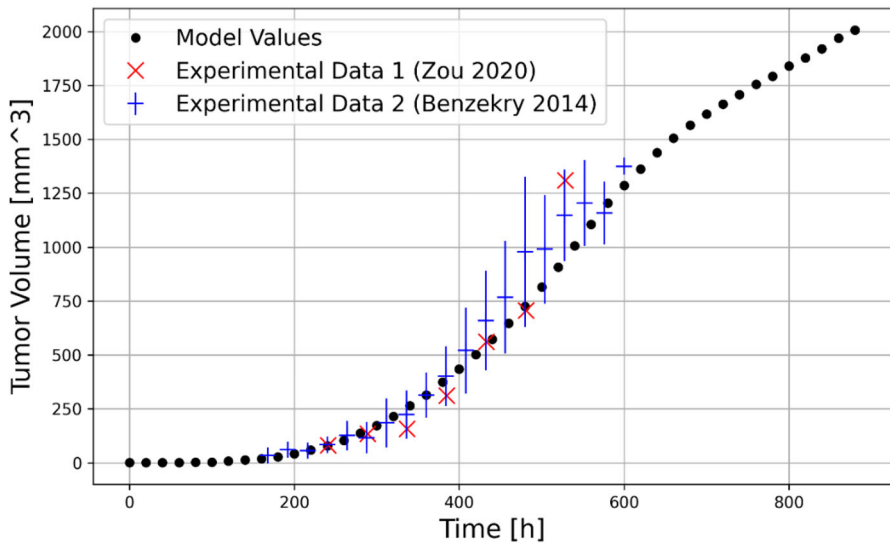


Fig. 6 Experimental Data against values generated using AMBER, matching the first data point to a data point from the model and tuning cell cycle duration and oxygen supply parameters. Data 1: Subcutaneous injection of xenograft A549 lung tumor (NSCLC) (Zou et al. 2020; Kim et al. 2012). Data 2: Subcutaneous injection of syngeneic lung tumor model in C57BL/6 mouse model. Uncertainty bars are standard deviations. (Benzekry et al. 2019) (Color figure online)

the tumor influence. The healthy vasculature consists of straight vessels, which match in vivo images (Forster et al. 2017; Jain 2003). In the tumor region, the vessels had a more chaotic structure, similar to the in vivo tumor images (Forster et al. 2017; Jain 2003) and an avascular center (Seynhaeve and Hagen 2016), which is also replicated by the model's simulations. This avascular center leads to a necrotic core spontaneously appearing in the simulation.

The metrics reported in Table 3 describe the vasculature shown in Fig. 7. Vessel length density is the total length of vessel per unit of volume and vessel segment length is the length of vessel between two branching nodes. As microvessels are not generated explicitly, the vessel length density is corrected to include the microvessels by adding the length of the implicit microvessels to the length of vessels in each voxel. Diameters and segment length are not corrected for the microvessels. Table 3 presents a comparison between the metrics we described and those from other models, along with metrics from imaging data derived from mice models.

3.5 In Silico Clonogenic Assays Accurately Replicate Irradiation Data

In order to test the rules of response to radiation assigned by the model in terms of cell damage and damage repair (see Supplementary Material), we set simulations independent of the rest of the model where arrays of isolated cells receive different radiation doses. The question was whether the incorporated damage induction and repair model was able to reproduce the behavior of in vitro clonogenic assay data

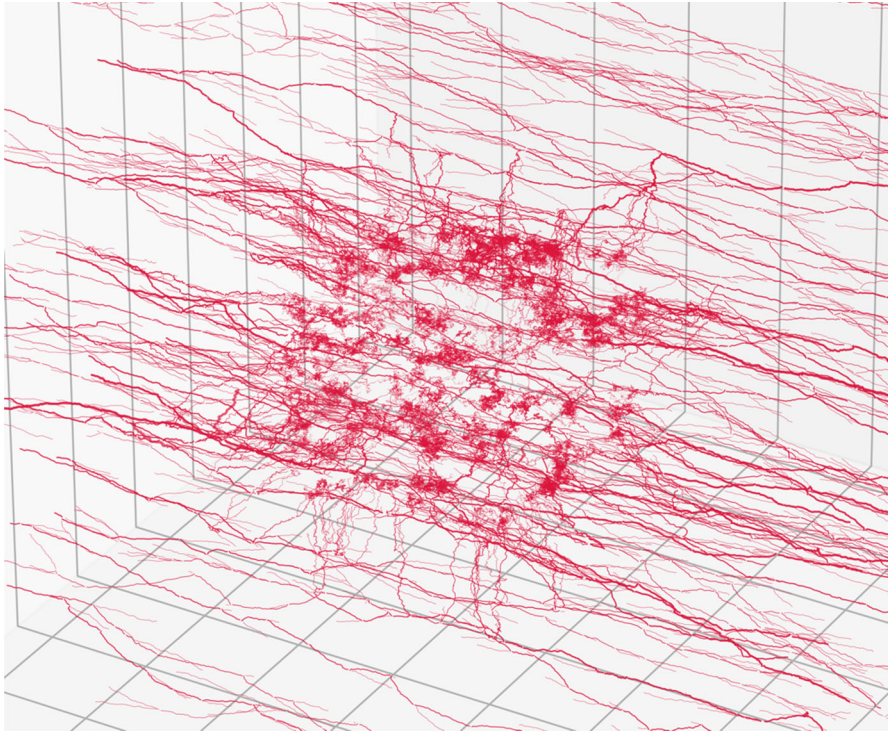


Fig. 7 Vasculature generated by the model for a tumor originating at the center of the domain, starting from a normal-like vasculature consisting of parallel blood vessels. Images of healthy vasculature and chaotic tumor vasculature for comparison can be found in refs. (Forster et al. 2017) and (Jain 2003), as well as in high-resolution intravital microscopy studies of tumors and associated vasculature (Seynhaeve and Hagen 2016). The simulation shows an avascular center matching in vivo data (Color figure online)

(Jiang et al. 2013). In this set of simulations, we quantify the viable fraction of cells after radiation as a function of the dose for different radiosensitivity values, i.e., the amount of damage caused to a cell per unit dose, and for different repair rates. Plotting the results of these simulations against the results from in vitro experiments shows that the damage induction and repair model effectively reproduces the behavior of the data (Fig. 8). Of note, it is well known that the data from clonogenic assays follows the LQ model (McMahon 2018), which here emerges spontaneously from the basic modeling rules without any explicit implementation. Altogether, these results confirm that the implemented rules of radiation response accurately resemble the behavior of the available biological knowledge.

3.6 Tumor Irradiation Achieves Tumor Control for Sufficient Dose per Fraction

In AMBER whole tumor simulations, we applied RT consisting of 5 cycles with 5 fractions per cycle. For this, TOPAS was used to calculate the dose delivered at each voxel by two opposing 6 MV photon beams and the damage induction and repair

Table 3 Metrics computed from the vasculature generated with AMBER for an unspecific tumor growth. Comparison with same metrics from other models and with experimental literature once the tumor growth reaches a steady state. Values are presented as min–max (average)

Ref	Computational			Experimental		
	AMBER	Nikmaneshi et al. (2020)	Norton and Popel (2016)	Stamatelos et al. (2014)	Gaustad et al. (2017)	Kim et al. (2012)
Diameter [μm]	4–71 (28)	5–65		10–60	7–14	8–64
Vessel Length Density [mm/mm ³]	0–281 (44)		8–234 (130)	0–250		9–101
Vessel Segment Length [mm]	0–3.65 (0.83)	0.05–0.56 (0.36)	0.07–0.23 (0.17)		0.8–1.60	0.13–0.43

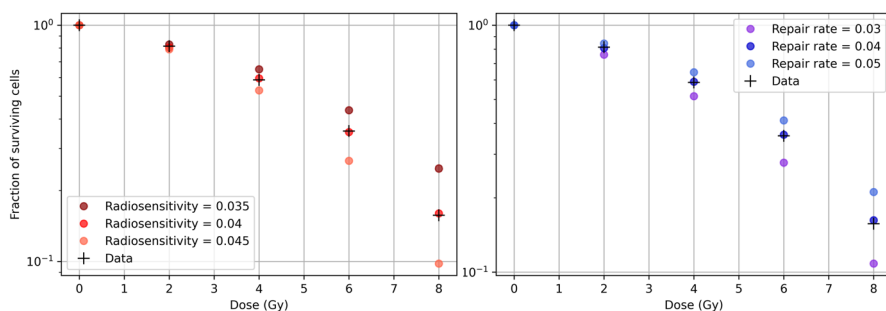


Fig. 8 In silico clonogenic assays for three different intrinsic radiosensitivity and three different repair rates (colored dots) showing survival fraction against applied dose and compared to in vitro clonogenic assays data (black crosses) from ref. (Jiang et al. 2013) (Color figure online)

model tracked the dynamic evolution of damage (Fig. SM1). Figure 9 shows the time evolution of the number of cells before, during, and after the treatment. The left panels show two scenarios for disparate values of the applied dose. From the results, it is evident that increasing the dose leads to tumor control, while lower dose leads to tumor relapse. We explored how the cell cycle duration affects survival by simulating the same treatment protocol under three different average cell duplication times. On the right panel of Fig. 9 it is shown that tumor control depends on the proliferation rate of the cells. Tumors with slower cycling cells are more likely to be controlled by the treatment, while faster cycling cells are associated with regrowth after irradiation for the same dose per fraction.

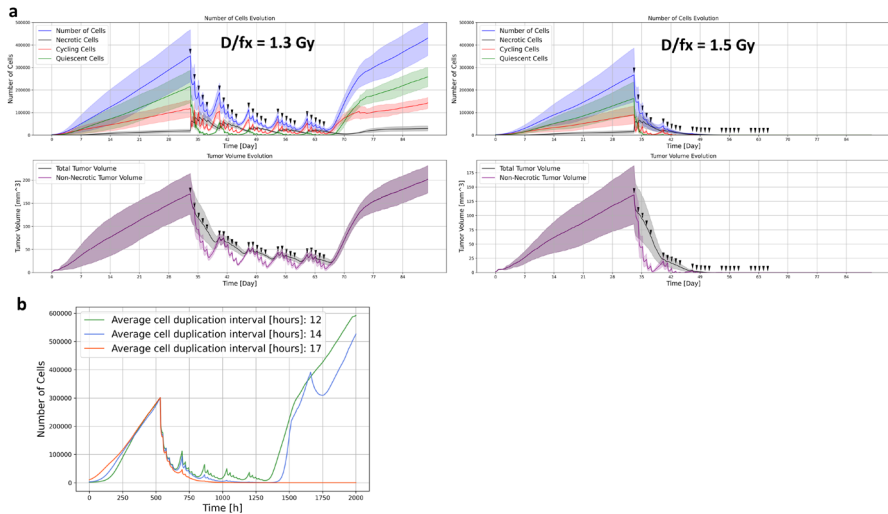


Fig. 9 **a** Evolution of the number of cells and tumor volume before, during and after irradiation. The tumor is irradiated with a 6MV photon beam using five times five fractions. Each fraction is separated by 24 h and each set of fractions is separated by a weekend. **a** Comparison of mean and standard deviation for 15 simulations with an average cell duplication time of 17 h under 1.3 Gy per fraction (left) and 1.5 Gy (right). **b** Effect of faster dividing cells on the tumor relapse for a dose per fraction of 1.5 Gy (Color figure online)

4 Discussion

In this article, we present AMBER, a framework for tumor growth and response to radiotherapy, which facilitates comprehensive macroscopic tumor modeling by enabling the integration of sub-scale models. To test our results, we compared tumor size curves and growth rate predictions with experimental measurements of growth of subcutaneous murine lung tumors. By adjusting oxygen supply and intrinsic cell doubling time, we successfully matched our simulations with experimental data. We showed that the computational tumor dynamics in our framework are compatible with classical mathematical laws of tumor growth, demonstrating that our approach follows similar behavior. This aspect is highlighted by the fact that the Gompertz and Holling model, closer to real tumor growth than the logistic model, are the ones that are the best matched with the behavior of AMBER. Additionally, we qualitatively compared the vasculature to experimental data and to other models, again showing agreement.

Using ABMs to investigate radiobiological effects presents several advantages compared to dynamics system models employing algebraic-differential equations. First, ABMs excel in representing heterogeneity by breaking down complex systems into individual components and their interaction rules. Secondly, ABMs demonstrate flexibility by allowing for the addition, modification, or removal of agents, as well as alterations to interaction rules, without necessitating adjustments to other agents and rules. Thirdly, ABMs offer enhanced interpretability, as one can modify the properties of an agent or an interaction and observe the direct consequences, e.g., on tumor growth (Ghaffarizadeh et al. 2018; Metzcar et al. 2019). Despite these strengths, low-abstraction ABMs are not very scalable, as computation times increase exponentially

when more details are added to the model. As a compromise, hybrid solutions that include both agent-based and continuum-like aspects can be utilized (Nikmaneshi et al. 2021; Ortega-Sabater et al. 2022; Bosque et al. 2023; Macklin et al. 2009; Cai et al. 2011; Wu et al. 2013; Kempf et al. 2015). Several hybrid ABMs of tumor growth have been published (see Table 1). However, most of them are focused on systemic therapies and offer limited functionality for radiation therapy. The models including radiation usually include insufficient components, e.g. missing the vasculature, to accurately model the effects of radiation on the tumor and the effect of radiation on the cell is usually simplified using the LQ model (O’Neil 2012; Brüningk et al. 2019). To accurately represent what is happening at the tumor scale many different scales must be considered, such as molecular interactions of reactive species with biomolecules, DNA repair and physical pressure gradients in the tissue. To the best of our knowledge, the only work approaching this was done by Liu et al. by bridging MC radiation delivery and an agent-based tumor growth model (Liu et al. 2021). However, this model focuses on the submillimeter scale, insufficient to model the behavior of a whole tumor and needs to be scaled up. In parallel, the approach used by Jiménez-Sánchez et al. (2021) and also suggested by Bravo et al. (Bravo et al. 2020), shows an alternative macro approach to hybrid ABMs well-fitting with the upscale requirements of a tumor growth model, but does not include radiotherapy or vasculature. To our knowledge, our present framework is the first work trying to bridge this gap.

Our results of radiation response in the simulation of *in vitro* assays, based exclusively on repair and vitality rules, showed a very good agreement with the data from clonogenic experiments, and consequently with the LQ behavior of those experimental data, which emerged spontaneously. Unlike analytical models, our approach provides additional information about the state and history of each cell and allows to change cell-specific properties and observe emergent changes. Furthermore, when a sample tumor is irradiated following a conventional treatment method of twenty-five fractions, the framework predicts a behavior compatible with radiation biology fundamentals and pre-clinical and clinical evidence. While sufficient dose leads to tumor control, a late relapse after an initial tumor suppression is observed for inferior doses. Notably, the effectiveness of tumor control is also linked to the rate of cell duplication, as highlighted in Fig. 9. Together, these results show the capability of our framework to reproduce coherent emerging behavior for the growth of a tumor to a pre-clinical volume and its exposure to radiation. A limitation of our approach is assuming that the dose delivered to each voxel is distributed homogeneously within the voxel, which will require further expansions to consider cases in which stochastic effects are necessary, such as the case of low radiation doses.

Despite the promising perspectives, there are limitations that need to be addressed. The current version of AMBER is an early-stage framework, and its predictive power is still limited, providing primarily qualitative results. For the moment, the framework lacks experimental validation for the parameter values, as it has not yet been extensively connected to datasets providing a quantitative description of specific cases. In particular, the model relies on several reference probabilities that influence the processes, which have been assigned reasonable values based on educated guesses but have not yet been calibrated to quantitative data. Future work will focus on connecting the model with data, allowing the current framework to be constrained to a specific

case. This will enable the provision of comprehensive quantitative insights, thereby extending the current applicability of the model, for the particular case considered. Several key features are also not yet included in the framework. Some characteristics of the tumor that are very simplified in the current implementation are the physical pressure, the cell cycle evolution of the tumor cells, the cell vitality function, and nuances of DNA repair. Moreover, the presence of non-cancer cells and the extracellular matrix, which can affect how cancer cells proliferate and migrate, have been neglected here. The current implementation of the models for these processes are placeholders and are destined to be replaced by better models. One key feature that is yet not included in the framework is the immune system, which holds a crucial role in radiation response. Other missing features include crosstalk from stromal cells, senescent cells, fibroblasts, and fibrosis (Piersma et al. 2020; Benzekry et al. 2019).

To improve the already existing components of the framework, one should take advantage of other ABMs to create better-accelerated models. For example, the models detailed in Table 1 and Nikmaneshi et al. (2020) or Vavourakis et al. (2017) for vasculature growth, Bekker et al. for immune infiltration (Bekker et al. 2022), Bravo et al. or Jiménez-Sánchez et al. for cell bunch migration (Jiménez-Sánchez et al. 2021; Bravo et al. 2020) and Liu et al. for the effect of radiation at the voxel scale (Liu et al. 2021) as well as the many open-source ABM frameworks such as PhysiCell, CompuCell, HAL and CHASTE. Concerning radiation, the use of more accurate models to describe DNA damage (Schuemann et al. 2018; Bertolet et al. 2023; Santos et al. 2014), DNA repair (Ingram et al. 2019; Henthorn et al. 2019, 2018, 2017; McMahon et al. 2017, 2016; Schuemann et al. 2018), and the effect of non-repaired lesions on the cells can be added to AMBER, for instance, via simulating radiation down to individual DNA strand breaks when using TOPAS-nBio (Schuemann et al. 2018). A potential extension of our framework is the inclusion of specific models for radiation effects on vessels (Wijerathne et al. 2021; Nikmaneshi et al. 2023). The integration of the multiple complementing advantages that these models present within AMBER will improve the radiotherapy-focused framework that we present here.

In our approach, an accelerated model can include more or fewer parameters depending on the accuracy of the model used to infer it. Different applications might require different versions of the accelerated model with more or less details depending on the application. The modularity of the framework allowed by the ABM implementation offers the possibility of implementing multiple models in parallel, from which users could choose. In its current version, AMBER includes simple models for cell death, oxygen delivery, cell migration, cell cycle, VEGF secretion, vessel growth, vessel maturation, and radiation damage. Connecting our framework to experimental data will constrain the model and its parameter values, allowing us to tailor general results to address specific problems. Several methods can be employed to parameterize the model, assuming adequate data is available. These methods can be broadly categorized into neural networks, generic optimization techniques, and Bayesian inference (Cogno et al. 2024). Bayesian inference is particularly well-suited to our modeling framework, as it offers a structured approach to estimate parameter distributions rather than relying on point estimates. This approach better captures the inherent uncertainty in biological systems and is flexible enough to accommodate various types of data, including those that are sparse and noisy. Specifically, Approximate Bayesian Computation (ABC) has

proven effective in similar contexts (Jiménez-Sánchez et al. 2021; Carr et al. 2021; Jørgensen et al. 2022; Wang et al. 2024). ABC involves iteratively simulating data under different parameter values and comparing the results with empirical data to identify those parameters that produce outputs closely resembling observed data. This method is particularly useful for complex stochastic models with many variables and can adapt to different types of sparse, noisy data. While ABC seems promising for identifying the model's parameters, the choice of method will ultimately depend on the type and quality of available data, and other techniques may also be necessary. Successfully parameterizing the model to fit specific data sets will enable us to derive meaningful insights from in silico simulations. Therefore, our framework allows modular user-specific modifications. Together, the observed emerging behavior compatible with known behavior, the implementation allowing easy extension and improvement, the hybrid approach allowing tumor of pre-clinical size, and the close interface with TOPAS and TOPAS-nBio allowing accurate radiation scoring at any scale show the large potential of AMBER.

5 Conclusion

In this article, we introduced AMBER, a framework developed to simulate the spatiotemporal growth of vascularized tumors and their response to radiation. Our current version allows us to simulate tumor growth and necrosis evolution and response to radiotherapy following a set of mechanistic rules for cell proliferation, cell migration, oxygen distribution, angiogenesis, cell death, and the effect of radiation on cells/vessels. This proof of concept was qualitatively compared with other computational and analytical models and murine experimental data, which confirmed the underlying logic of the framework and its capability to lead to a coherent emerging behavior. To improve the framework's accuracy, experimental validation of the parameters and improvements of existing components of the framework, as well as the introduction of additional components, are essential steps. Despite the need for further improvement, the possibilities for future applications are vast, and AMBER is the first step in our pipeline to model radiation effects from basic principles up to the tumor scale.

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